



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

NTU Academy for Professional
and Continuing Education

n8n basics

Visual No-Code Building Platform

Module 6

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01

Origins

Why n8n exists, and why that matters for what you will build

n8n was built by a VFX artist tired of repeating himself

Jan Oberhauser

Berlin, 2019. Worked on Maleficent and Happy Feet Two before he got tired of writing the same glue code over and over. Built a tool to wire software together visually.

The name

"nodemation" was too long. Eight letters between the two n's → n8n.
Pronounced n-eight-n.

2026 today

230,000+ users · 3,000+ enterprise customers · \$2.5B valuation after the Oct 2025 Series C.

n8n

A fair-code workflow automation platform that empowers users to connect and automate anything—from simple tasks to complex systems.

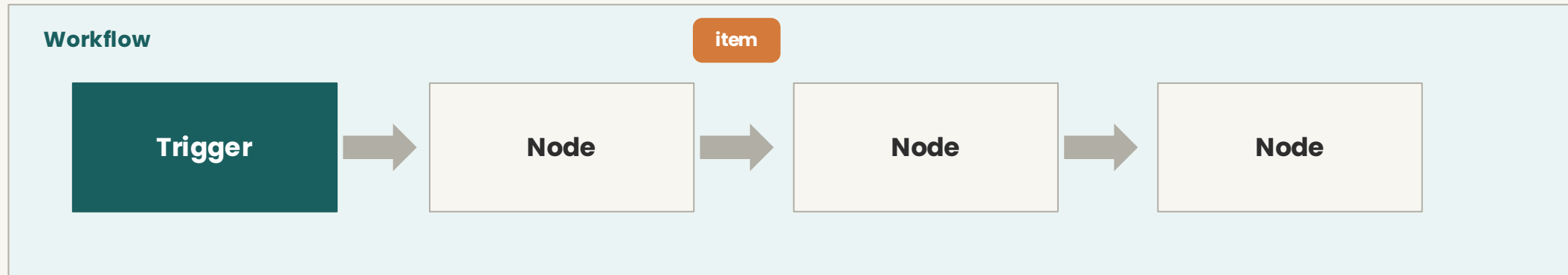
02

Fundamentals

The five ideas that explain every workflow

"A workflow is a chain of nodes that pass data from one to the next, starting with a trigger."

Every workflow has the same five-part anatomy



1 Workflow

The whole automation — active or inactive.

2 Nodes

Boxes on the canvas. Each does one job.

3 Trigger

First node. No input. Starts the flow.

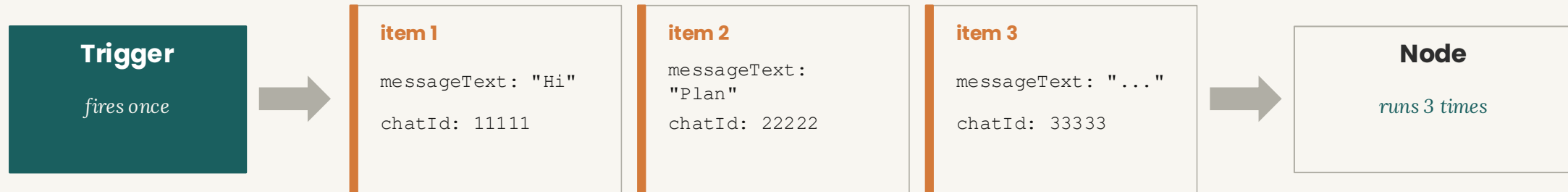
4 Connection

Arrows. Output of one → input of next.

5 Item

JSON package of data. Nodes run once per item.

Items are how data moves — each one a small JSON package



One trigger event → a list of items → the next node runs once per item

`one item, up close`

```
{ "json": { "messageText": "Hello", "from": "Swarup", "chatId": 12345 } }
```

Click any node after a run, see exactly what came in and what came out. This is how you debug.

Expressions pull data between nodes with a double-curly syntax

When you fill in a node parameter you have two choices: type a fixed value, or use an expression to pull a value from a previous node.

expression

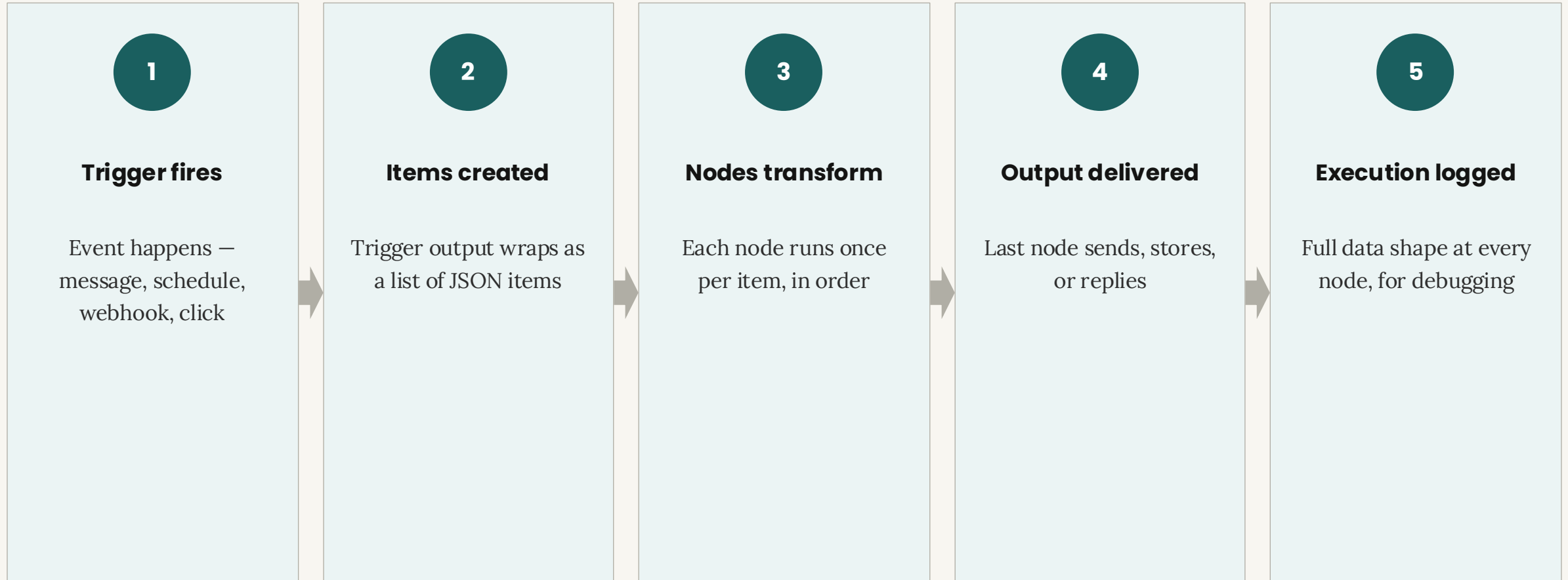
```
{{ $json.messageText }}
```

```
{{ $now }}
```

```
{{ $('Telegram Trigger').item.json.message.chat.id }}
```

In practice, you almost never type these by hand. n8n has a click-to-pick UI — you click the field you want in a previous node, and the expression is generated for you.

n8n runs your workflow once per item — and logs every step



03

The 16 nodes

One slide to see them all, four to know them

Sixteen nodes, four families — the working AI workflow toolkit

Triggers — start a workflow (4)

Manual · Schedule · Webhook · Telegram Trigger

Flow — glue every workflow (5)

HTTP Request · Edit Fields (Set) · IF · Switch · Code

AI — the brain (6)

AI Agent · Basic LLM Chain · Chat Model · Simple Memory · Embeddings · Vector Store

Output — close the loop (1)

Telegram — Send a text message

Explore all 400+ n8n nodes → docs.n8n.io/integrations

Swarup Biswas | swarupbiswas.com



Triggers — four ways a workflow starts

Manual Trigger

Click "Test workflow" in the editor. Use it for development, debugging, and demos.

Schedule Trigger

Run on a clock — every minute, hour, day, or cron expression. Daily AI briefings live here.

Webhook

Generates a unique URL. Any external system that hits it pushes data in. The universal entry point.

Telegram Trigger

A specialised webhook for Telegram. Fires when your bot receives a message, photo, or command.

Flow — five universal glue nodes

HTTP Request

Calls any URL. The escape hatch for any service without a dedicated n8n node.

Edit Fields (Set)

Rename, add, remove, restructure fields between nodes. How you build a clean prompt input.

IF

Two-way fork. Confidence above 0.8? Send. Below? Route to human review.

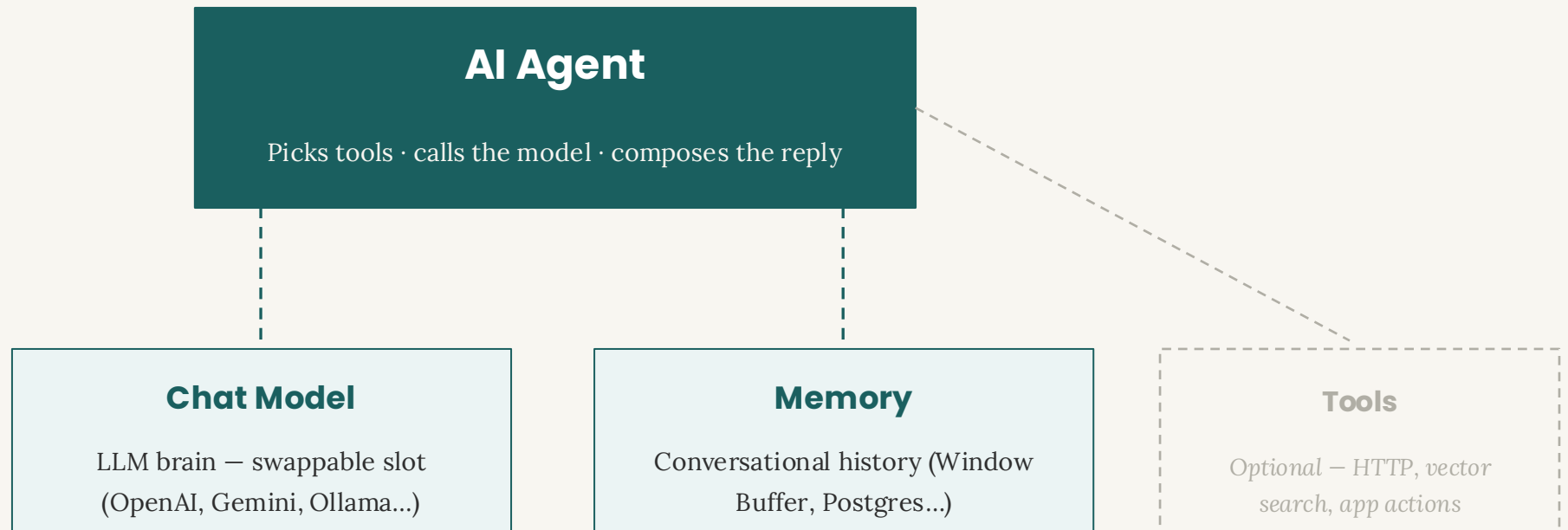
Switch

Multi-way fork. Intent = support → workflow A. Sales → workflow B. Feedback → workflow C.

Code

JavaScript (or Python in Cloud) escape hatch when no node fits. Use sparingly.

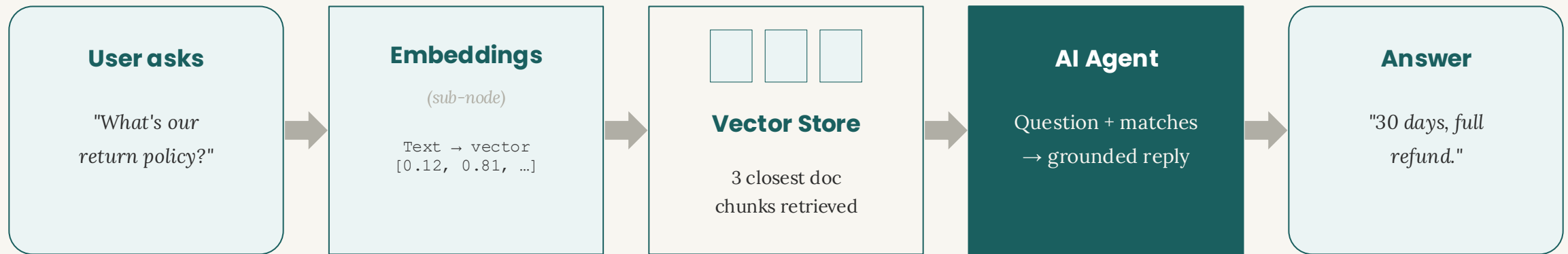
AI Agent is the hub — Chat Model and Memory plug in beneath



Basic LLM Chain

Simpler alternative. Prompt in, text out — no tool-picking, no decisions. Same sub-node slots for Chat Model and Memory. Use for translate, summarise, classify.

Embeddings + Vector Store = RAG — chat with your own docs



Same Agent pattern from the atomic bot — just give it Embeddings + Vector Store sub-nodes

Embeddings turn text into searchable vectors. Vector Store finds the closest matches. The Agent receives the question plus the retrieved chunks and answers from your own content — not its training data.

04

Environment

Where the workflow actually runs

Two paths to a running n8n — Cloud or self-hosted

n8n Cloud

- In your browser at app.n8n.cloud
- 14-day free trial, no card needed
- Public HTTPS URL for webhooks built in
- LLMs via paid APIs (Gemini, OpenAI, Claude)
- Today's session uses this path

Self-Hosted AI Starter Kit

- Docker on your own machine
- Free forever, uses your hardware
- Need tunnel mode for public webhooks
- Local LLMs via Ollama — fully offline
- Self-Hosted AI Starter Kit ships ready to go

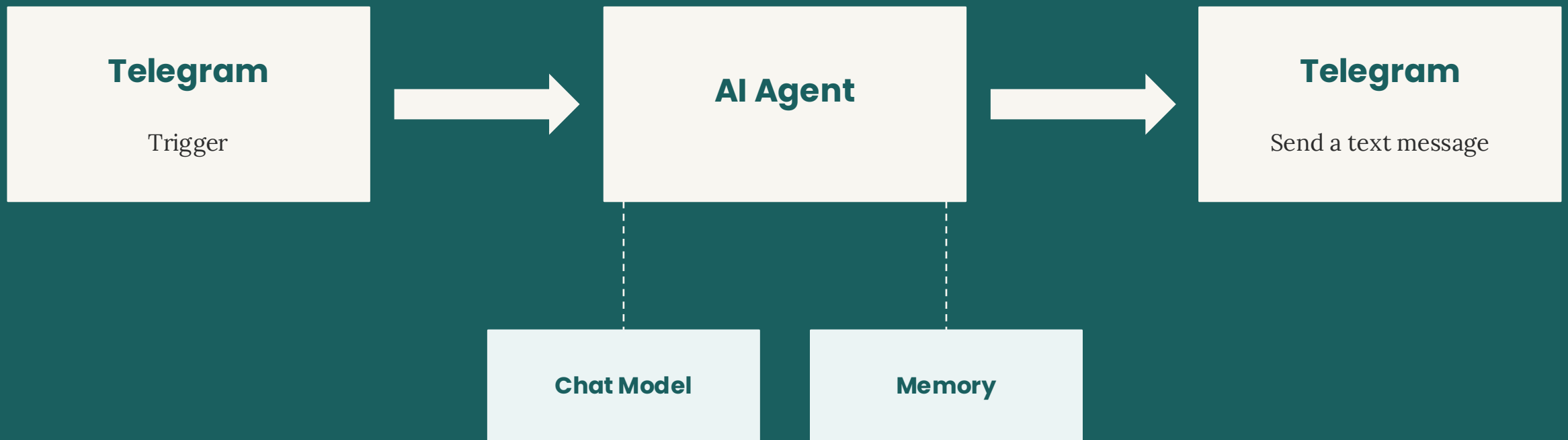
Workflows export as JSON. You can build in Cloud today, move to self-hosted tomorrow — no rebuild.

05

The atomic AI app

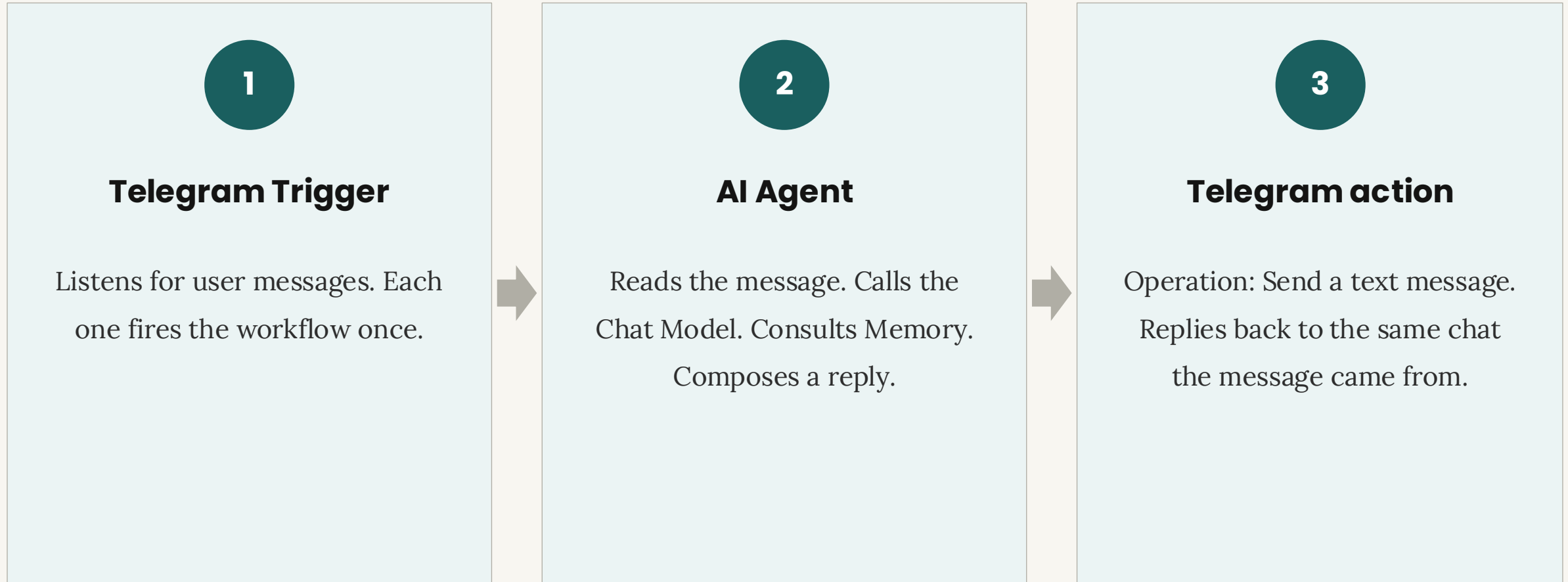
The smallest AI app there is — and the pattern behind every bigger one

"Three nodes are the entire bot."

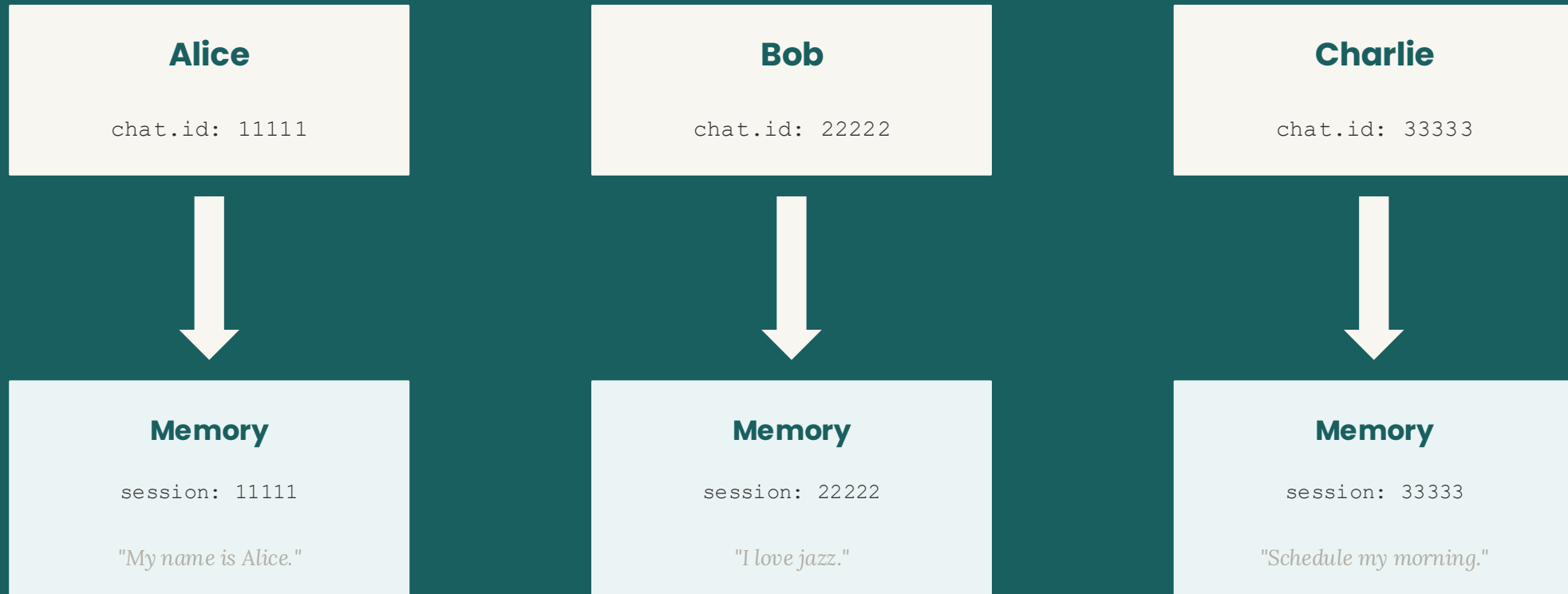


Two sub-nodes plug into the Agent — they hang beneath, not in the main chain

Trigger → AI Agent → Send a text message — the universal AI loop



Memory keyed on chat ID gives every user a separate history



```
Session ID = {{ $('Telegram Trigger').item.json.message.chat.id }}
```

Without this, all users share one memory and the bot gets confused.

Live demo

► Live demo

8 min

- Open the n8n workflow — three nodes already wired
- Show the Telegram Trigger's saved Bot credential, the AI Agent's System Message, and the Chat Model with Gemini selected
- Send "My name is [participant's name]" from a phone — bot replies in real time
- Send "What's my name?" — bot answers correctly
- Open Executions, click into the latest run, walk through items at each node

One extra node turns this loop into RAG, agents, or multi-channel bots

Add this	To get	New capability
Vector Store + Embeddings sub-nodes	Knowledge base on the Agent	RAG — chat with your own docs
HTTP Request as an Agent tool	Agent can take real actions	Books meetings, sends emails, queries APIs
Swap Telegram nodes for Slack / WhatsApp	Same bot, different surface	One brain, many channels
Switch node before the Agent	Route by intent or user tier	Free users → cheap model, paid → premium
Postgres Memory instead of Window Buffer	Memory that persists across sessions	Bot remembers users weeks later
Schedule Trigger instead of Telegram	Time-driven AI	Daily AI briefing pushed to your chat

Every more ambitious AI app is this same loop, plus one or two more nodes.

Version Management

Month/Year	Version	Changes
May 2026	0.0	New Module Design